Infringement Profile of AMG Petronas Mercedes F1 Team using Summarization Techniques

FINAL REPORT
Anushree Udhayakumar (au11), Pranavh Harish Sivakumar (phs4)

Abstract

Steward reports in Formula One contain important details about rule violations, but they are plenty and difficult to compare across seasons. This makes it hard to understand how a team's racing behavior changes over time. In this project, we collect five years of steward reports from the FIA website and analyze them to build an infringement profile for the AMG Mercedes F1 Team. We use both extractive and abstractive summarization methods, focusing on simple and interpretable models that make the patterns easy to understand. By doing this, we aim to highlight the most frequently occurring infringements and provide a clear, data-driven view of the team's on-track conduct.

1 Introduction

During a Formula One race, if any driver or team commits a violation of FIA policies or regulations, it is documented and published on the FIA's official website. These reports follow a consistent structure that describes the violation, the decision taken by the stewards, and any penalties imposed. We aim to identify patterns of violations in these reports and build a team-level infringement profile, where profiling refers to systematically aggregating and summarizing repeated violations over time to characterize a team's typical behaviors, tendencies, and areas of recurring concern. This profile can help improve and monitor the team's trajectory of rule violations across seasons.

This report presents a brief introduction to the dataset, followed by the pre-processing steps and descriptive analysis of the corpus. It then summarizes our experiments with multiple summarization models under different parameter settings and concludes with an interpretation of the results.

2 Dataset

The data is manually downloaded from the FIA official website. All team violation reports are downloaded after which during pre-processing we filter them for the AMG Petronas Formula One team. Link to dataset website: Decision documents | Federation Internationale de l'Automobile

3 Summary of tasks and goals

We began by manually downloading FIA infringement reports from 2020 to 2024. While doing this, we excluded the documents we did not need, such as meeting notes and pre- or post-race reports, so that only the actual infringement reports were kept for analysis. The preprocessing pipeline filtered for Mercedes-specific documents, extracted text from PDFs, and removed redundant and unnecessary parts (headers, footers, administrative metadata). The goal for this pre-processing was to produce clean text files without redundancy or irrelevant information.

After cleaning the data, we ran basic descriptive analyses to look for patterns in the violations. We examined how often each driver was involved, and which tracks saw the most incidents, common keywords in the reports, to get a sense of how these violations changed over the five-year period. This helped us understand what to expect when generating summaries.

Our intention was to test and compare TextRank, LexRank, DistilBART, and Pegasus algorithms. We chose these models to explore the spectrum

from extractive (TextRank, LexRank) to abstractive (DistilBART, Pegasus) summarization approaches, expecting different trade-offs between factual preservation and readability. The evaluation metrics we use are compression ratio, key-entity overlap, redundancy, ROUGE, and FactCC. These metrics assess how well each model condenses information, preserves important entities, avoids repetition, matches reference summaries, and maintains factual accuracy. In order to evaluate the generated summaries using ROUGE, we generated gold-reference summaries using a LLM-based approach with GPT-4o-mini and Groq LLaMA-3.3-70B, as manual annotation of 82 documents would require weeks.

Year-wise summaries are generated for every model (2020–2024), and an overall summary is also generated for each model. The overall summary should progressively improve from TextRank to LexRank to DistilBART to Pegasus, and we expect these metrics to reflect that progression, showing increasing sophistication in handling domain-specific regulatory text.

4 Literature Review

Several studies have explored summarization of sports-related documents, particularly live commentary and match reports. Huang et al. (2020) proposed neural approaches for generating sports news summaries from live commentary, highlighting challenges such as redundancy and noisy event streams. Puduppully et al. (2019) introduced **SPORTSSUM**, a large-scale dataset for generating soccer match reports from commentary and showed that abstractive models improve readability but struggle with factual grounding. Wiseman et al. (2017) studied data-to-text generation for sports, demonstrating that structured game statistics can be transformed into coherent narrative summaries, though maintaining factual accuracy remains difficult. More recently, Liu et al. (2021) incorporated external knowledge to improve entity and event consistency in sports game summarization, reinforcing the importance of grounding in entity-heavy domains.

Building on prior work in sports document summarization and behavioral analysis, we next draw from established research in extractive summarization to inform our methodological choices. The literature review directly informed us about our implementation of **TextRank and LexRank** for profiling Mercedes F1 infringements across multiple dimensions. Mihalcea and Tarau (2004) introduced TextRank as a graph-based ranking algorithm adapted from PageRank, demonstrating its effectiveness for unsupervised sentence extraction. This work provided the theoretical basis for our sentence similarity computation using TF-IDF vectorization and cosine similarity and motivated our use of graph centrality for sentence ranking.

The comparative study by Ghorpade et al. (2024) offered practical insights into the relative strengths and limitations of TextRank and LexRank across document types, guiding both our algorithm selection and our expectations of their performance on structured regulatory text. Additionally, Akhmetov et al. (2022) examined the upper bounds of ROUGE scores achievable by extractive summarization methods, which helped contextualize the lower ROUGE values observed when comparing extractive outputs against LLM-generated gold-standard summaries.

Further support for our approach comes from Yugandhar et al. (2023), who demonstrated TextRank's effectiveness in condensing large volumes of text while preserving key information, aligning closely with our use case of summarizing lengthy infringement reports. Finally, Givchi, Ramezani, and Baraani-Dastjerdi (2022) showed that graph-based summarization methods perform well on complex technical documents, reinforcing our choice of unsupervised, centrality-driven algorithms for domain-specific regulatory analysis.

5 Descriptive Statistics

The analysis reveals Mercedes F1 team's infringement patterns across 2020-2024 (Fig 1) through 82 total violation documents. Violations peaked in 2023 with 23 violations, while 2020 had the lowest at 11 violations, averaging 16.4 violations per year. Document length analysis (Fig 1) shows 2024 had the longest documents (1,862 characters average) and 2022 the shortest (1,498 characters), with an overall average of 1,673 characters per document.

Driver-specific analysis (Fig 2) reveals Car 44 had its most problematic year in 2020 with 10

violations, while Valtteri Bottas (Car 77) peaked in 2021 with 4 violations, and George Russell (Car 63) had 6 violations in 2023.

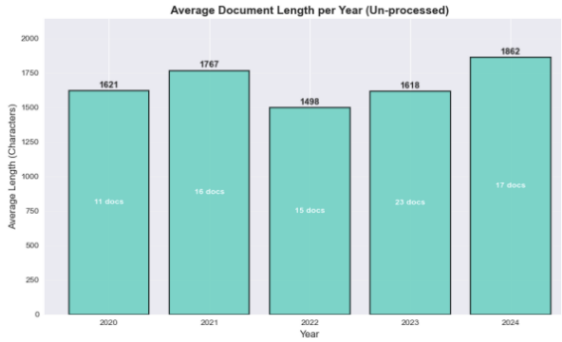


Fig 1: Average Document Length per Year (un-processed)

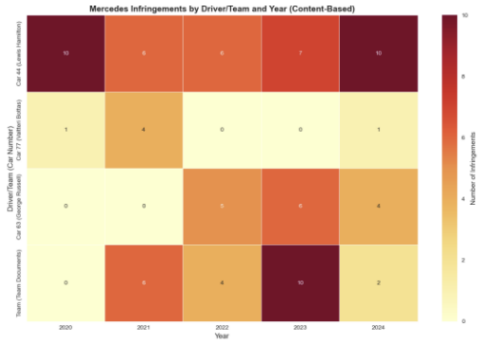


Fig 2: Infringements by Driver or Team by year

Keyword frequency analysis (Fig 3) reveals the nature of Mercedes F1's infringement patterns through common violation types. "Fined" appears most frequently (19 occurrences) because FIA regulations mandate financial penalties for procedural violations such as late attendance at press conferences, pre-race delegation failures, inappropriate radio communications, or post-race protocol breaches. "PU" violations (14 occurrences) reflect the complex power unit regulations where Mercedes must seek FIA approval for any changes, whether due to damage repairs or performance updates, often requiring impromptu decisions during race weekends. "Collision" incidents (13 occurrences) represent the most common driver errors, indicating that despite Mercedes' technical excellence, racing incidents remain inevitable in competitive Formula One environments, demonstrating that both technical violations and safety incidents dominate their infringement profile.

Track-wise analysis (Fig 4) identifies Austrian Grand Prix as the most problematic venue with multiple violations. This corroborates the fact that the Red Bull Ring's high-speed straights and flowing corners create conditions where drivers can push aerodynamic limits, leading to more frequent track limit violations as cars carry higher speeds through corners. Upon preprocessing (Fig 5), we reduced document length by approximately 50% (from 1,673 to 870 characters average, refer Fig 1), that is by removing administrative headers and footers while preserving core infringement content for text mining analysis.

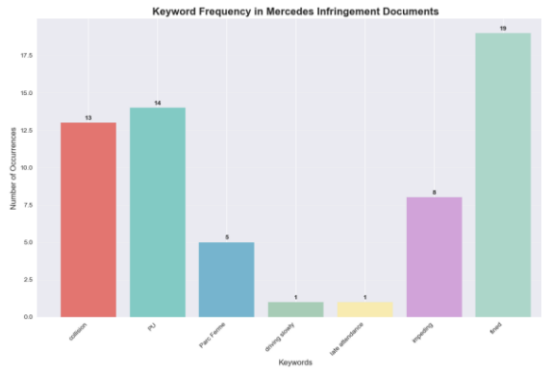


Fig 3: Keyword Frequency in corpus

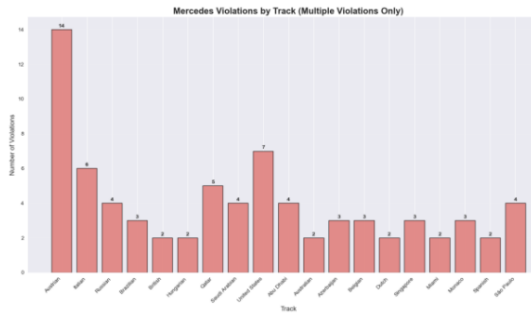


Fig 4: Violations by track

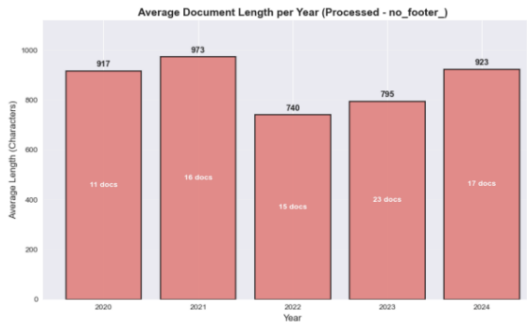
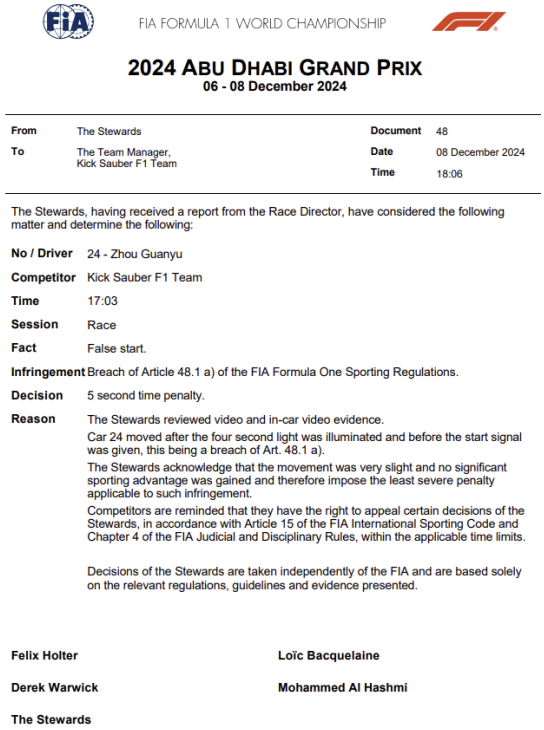


Fig 5: Average document length per year (pre-processed corpus)

6 Preprocessing

We systematically cleaned Mercedes F1 infringement documents through four progressive preprocessing steps. First, we removed common articles ("the", "a", "an") to reduce noise while



FIA FORMULA 1 WORLD CHAMPIONSHIP

2024 ABU DHABI GRAND PRIX
06 - 08 December 2024

From	The Stewards	**Document**	48
To	The Team Manager, Kick Sauber F1 Team	**Date**	08 December 2024
		Time	18:06

The Stewards, having received a report from the Race Director, have considered the following matter and determine the following:

No / Driver	24 - Zhou Guanyu
Competitor	Kick Sauber F1 Team
Time	17:03
Session	Race
Fact	False start.
Infringement	Breach of Article 48.1 a) of the FIA Formula One Sporting Regulations.
Decision	5 second time penalty.
Reason	The Stewards reviewed video and in-car video evidence. Car 24 moved after the four second light was illuminated and before the start signal was given, this being a breach of Art. 48.1 a). The Stewards acknowledge that the movement was very slight and no significant sporting advantage was gained and therefore impose the least severe penalty applicable to such infringement. Competitors are reminded that they have the right to appeal certain decisions of the Stewards, in accordance with Article 15 of the FIA International Sporting Code and Chapter 4 of the FIA Judicial and Disciplinary Rules, within the applicable time limits. Decisions of the Stewards are taken independently of the FIA and are based solely on the relevant regulations, guidelines and evidence presented.

Felix Holter **Loïc Bacquelaine**

Derek Warwick **Mohammed Al Hashmi**

The Stewards

Fig 6: Unprocessed infringement report by FIA

preserving meaningful content, achieving a 7.1% text reduction. But this step was undone after preliminary summary generation because the summaries were difficult to read without articles. Second, we eliminated FIA document headers containing administrative information like "From/To", document numbers, dates, and steward details by finding the "No / Driver" keyword and removing everything before it. Third, we removed competitor and time information patterns like "Competitor Mercedes-AMG Petronas F1 Team Time 15:59" using regex to strip administrative 205 metadata. Finally, we removed appeal information and signatures starting from "Competitors are reminded..." to eliminate legal footers and steward signatures. Each step created progressively cleaner files with prefixes "no_articles_", "no_header_", "no_comp_time_", and "no_footer_" respectively, ultimately producing 82 clean documents ready for text mining and summarization analysis.

7 Generating Gold-standard summaries

Due to the unavailability of gold-standard summaries, we resorted to using LLMs to generate what we treat as gold standards. To objectively evaluate summarization model performance, we employed a controlled, LLM-assisted hierarchical (or step-by-step) summarization approach. The resulting reference summaries were then used as benchmarks for computing ROUGE and factual consistency metrics against model outputs. All 82 preprocessed Mercedes infringement reports (2020 - 2024) were processed through a three-stage hierarchical pipeline. At Level 1, reports were grouped into chunks of eight documents and summarized into approximately 100-word summaries. To reduce hallucinations, verified infringement counts were injected directly from the source data rather than inferred from text. At Level 2, chunk-level summaries were aggregated into yearly summaries (~200 words), capturing infraction distributions, penalty trends, driver-specific patterns, and temporal trends. At Level 3, yearly summaries were combined into a single multi-year profile (200–250 words), with global infraction totals enforced to maintain numerical consistency across the corpus. Gold-standard summaries were generated using **GPT-4o-mini** and **Groq LLaMA-3.3-70B-Versatile**, using identical prompts and low-temperature settings (0.1–0.2) to ensure reproducibility and minimize variability. Since we are using LLMs, we also evaluated the summaries it produced just like every other summary in our experimentation, and the results are consolidated in table 1.

	Compression ratio	Redundancy check	Key-Entity Overlap	ROUGE-1 (0.3–0.5)	ROUGE-2 (0.05–0.20)	ROUGE-L (0.2–0.4)
GPT-4o-mini	98.09%	7.77	64.29%	0.5973	0.2400	0.2788
Groq-Llama	98.30%	9.42	66.67%	0.5973	0.2400	0.2788

Table 1: Evaluation of Gold-standard summaries

8 Experimentation

8.1 Models

We selected four models to systematically explore both extractive and abstractive summarization approaches. Our extractive baselines, TextRank and LexRank, operate by selecting sentences directly from the source document. TextRank Mihalcea & Tarau (2004) uses a graph-based ranking mechanism where sentences form nodes connected by similarity edges, and importance is derived through iterative score propagation. LexRank Erkan & Radev (2004) similarly builds a sentence similarity graph but leverages eigenvector centrality to identify the most representative sentences in the document.

For abstractive summarization, we employed **DistilBART** and **Pegasus**, both transformer-based sequence-to-sequence models that generate novel text rather than extract existing sentences. DistilBART (Lewis et al., 2020, Alaei, S. 2023) is

a compact, distilled variant of BART that maintains strong generative capabilities while being computationally efficient. Pegasus (Zhang et al., 2020), pre-trained with a gap-sentence-generation objective, is specifically optimized for summarization. Its pre-training on C4 - a 750GB cleaned, diverse corpus - and HugeNews with 1.5B news sentences equips it with robust knowledge of factual reporting, event structure, and journalistic style, all highly aligned with FIA infringement document patterns.

Together, these models allow a clear comparison between extractive methods that preserve original phrasing and abstractive models that synthesize new language, enabling us to evaluate how summarization complexity and generative depth influence summary quality for infringement narratives.

8.2 Parameters

The extractive algorithms share four parameters: (1) Sentence Count, which defines the number of top-ranked sentences to extract for the summary, controlling summary length; (2) Similarity Threshold, which defines the minimum cosine similarity between sentences to create graph edges, determining when sentences are considered similar enough to connect in the similarity graph; (3) Minimum Text Length, which defines the minimum character length required for summarization, filtering out documents that are too short to summarize effectively; (4) PageRank Alpha, specific to TextRank, is the damping factor for PageRank, controlling the probability of following graph edges versus random jumps in the PageRank algorithm; and (5) Diversity Threshold, specific to LexRank, is the similarity threshold for ensuring sentence diversity, preventing the selection of redundant sentences that are too similar to already selected ones.
For the abstractive algorithm Pegasus, the parameters are (1) Maximum Length, which defines the maximum number of tokens in the generated summary, limiting the upper bound of summary size; and (2) Minimum Length, which defines the minimum number of tokens in the generated summary, ensuring a minimum output length. All parameter settings-experiments are summaries in table 2.

Model	Parameter	Value
Extractive Models		
TextRank	(Sentence Count, Similarity Threshold, PageRank Alpha, Minimum Text Length)	TextRank_v1:(5, 0.1, 0.85, 100) TextRank_v2: (8, 0.15, 0.85, 100)
LexRank	(Sentence Count, Similarity Threshold, Diversity Threshold, Minimum Text Length)	LexRank_v1: (5, 0.1, 0.7, 100) LexRank_v2: (8, 0.15, 0.7, 100)
Abstractive Models		
DistilBart	(Maximum Length, Minimum Length, Maximum Input Length)	DistilBart_v1: (512, 50, 1024) DistilBart_v2: (768, 100, 1024)
Pegasus	(Maximum Length, Minimum Length, Maximum Input Length)	Pegasus_v1: (512, 50, 1024) Pegasus_v2: (768, 100, 1024)

Table 2: Experimentational setup

9 Evaluation and Interpretation

To evaluate, we used straight forward metric - ROUGE and content-quality metrics such as key-entity overlap, redundancy, compression, and factual consistency. Together, we assess not only how similar a summary is to a reference, but also how informative, concise, and factually accurate it is. ROUGE (Recall-Oriented Understudy for Gisting Evaluation) measures lexical overlap between a generated summary and a reference summary and not the corpus (Lin, 2004). We used:

- ROUGE-1 measures unigram (word-level) overlap and reflects how much of the main content is shared.
- ROUGE-2 measures bigram overlaps and captures short phrase similarity.
- ROUGE-L measures the longest common subsequence, which reflects similarity in sentence structure and ordering.

Key-Entity Overlap measures how many important named entities or key concepts from the original document are preserved in the summary. Unlike ROUGE, it does *not* depend on matching a reference summary. Instead, it evaluates whether the summary retains the critical informational elements of the source text. Next, we found **Redundancy score** which measures repetition within the generated summary itself**.** It quantifies how often the same words, phrases, or sentences are repeated internally. This metric does *not* compare the summary to the source document or corpus; it strictly evaluates internal repetition, where lower scores indicate more concise summaries. **Compression Ratio** measures how much shorter the summary is compared to the original document. And finally, **FactCC** evaluates factual consistency between the summary and the source document**,** identifying whether the

summary introduces unsupported or incorrect statements (Kryściński et al., 2020).

9.1 ROUGE-Based Evaluation

We first evaluated all models using ROUGE scores against two gold-standard summaries generated by GPT-4o-mini and Groq-LLaMA. To contextualize these scores, we compared them against typical ROUGE ranges reported for strong summarization systems such as PEGASUS, BART, T5, and Pointer-Generator Networks (Zhang et al., 2020; Lewis et al., 2020; Raffel et al., 2020; See et al., 2017). Across both gold standards, DistilBART achieved the highest ROUGE-1 and ROUGE-L scores, indicating that it best retains core content and preserves overall structural similarity with the reference summaries. Refer Table 3. This makes DistilBART the strongest model in terms of broad semantic alignment. While Pegasus did not outperform DistilBART on ROUGE-1 or ROUGE-L, Pegasus_v2 achieved the highest ROUGE-2 scores under both evaluation settings. This suggests that Pegasus is particularly effective at matching short multi-word phrases and local phrasing patterns, even though it does not perform as strongly on global content coverage or structure.

GROUND TRUTH	METHOD	ROUGE-1 (0.3–0.5)	ROUGE-2 (0.05–0.20)	ROUGE-L (0.2–0.4)
gpt-4o-mini	TextRank	0.209	0.034	0.124
	TextRank_v2	0.209	0.025	0.138
	LexRank	0.248	0.038	0.137
	LexRank_v2	0.257	0.004	0.128
	DistilBart	***0.330***	***0.035***	***0.165***
	DistilBart_v2	0.302	0.029	0.140
	Pegasus	0.293	0.012	0.145
	Pegasus_v2	0.268	0.047	0.134
groq-llama	TextRank	0.181	0.018	0.133
	TextRank_v2	0.188	0.013	0.131
	LexRank	0.227	0.013	0.133
	LexRank_v2	0.256	0.009	0.130
	DistilBart	***0.302***	***0.021***	***0.143***
	DistilBart_v2	0.260	0.017	0.119
	Pegasus	0.265	0.012	0.118
	Pegasus_v2	0.262	0.038	0.137

Table 3: Summary of ROUGE evaluation

9.2 Evaluation Using Content-Quality Metrics

Among extractive models (TextRank and LexRank), we observe 100% key-entity overlap, which is expected since these methods directly copy sentences from the source document. However, this comes at a significant cost. Extractive models exhibit very high redundancy, indicating frequent repetition of similar sentences, and they show lower factual consistency scores, suggesting weaker alignment with the document's factual intent. In other words, while extractive methods preserve entities extremely well, they often repeat information and include unnecessary text, resulting in less concise summaries.

Among abstractive models, performance is more balanced. DistilBART achieves 89% key-entity overlap, very low redundancy (6.32), and the highest FactCC score (0.9983), indicating that it preserves important information while minimizing repetition and maintaining strong factual accuracy. Pegasus performs similarly, but it never surpasses DistilBART overall. In fact, Pegasus_v2 exhibits higher redundancy (8.25) and more aggressive compression, which slightly reduces clarity despite its strengths in phrase-level matching. All these values are summarized in table 4.

	Key- Entity Overlap	Compression Ratio	Redundancy check	FactCC
TextRank	100%	99.02%	60.62	0.8842
TextRank_v2	69.57%	97.78%	39.78	0.99
LexRank	100%	99.37%	7.72	0.7537
LexRank_v2	80.00%	97.79%	6.90	0.995
DistilBart	***89.47%***	***98.72%***	***6.32***	***0.9983***
DistilBart_v2	71.43%	96.79%	5.39	0.988
Pegasus	81.82%	97.75%	5.43	0.9318
Pegasus_v2	71.43%	98.85%	8.25	0.997

Table 4: Summary of evaluation using Content-Quality Metrics

Taken together, these results show that DistilBART is the best all-around model for summarizing FIA steward reports. It achieves the strongest balance between content retention, structural similarity, conciseness, and factual consistency. Pegasus_v2 contributes complementary strengths, particularly in capturing bigram-level phrasing as reflected in ROUGE-2, but it does not match DistilBART's overall reliability. These findings highlight how different summarization approaches behave in highly formal, compact, and domain-specific documents, and they guide the selection of models best suited for producing accurate and readable summaries.

10 Limitations

Although DistilBART produced readable and factually accurate summaries that covered the most frequent violations, the outputs lacked a

strong narrative structure. Team and driver names were often repeated, leading to increased redundancy. In addition, the summaries did not fully achieve team-level behavioral profiling; instead, they resembled enhanced extractive summaries that primarily stitched together individual incidents with improved filler words.

These results highlight the limitations of summarization models we experimented with, when applied to highly formal, monotonous and structured reports. While abstractive models improve readability and factual consistency, they still struggle to generate interpretive, "narrative-driven profiles" without additional guidance.

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